

Market Regime Detection and the Robustness of Simple Allocation Strategies

A reproducible walk-forward study (RegimeLab, v3)

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Abstract

We study whether market regime detection improves the *robustness* — rather than the headline performance — of simple allocation strategies, out-of-sample and net of transaction costs. Using a walk-forward framework on the S&P 500 ETF (SPY, 2005–2026) with proportional costs and a T-bill cash leg, we compare buy-and-hold and trend-following baselines against variants gated by a transparent volatility filter and by 2- and 3-state Gaussian hidden Markov models (HMMs) restricted to *filtered* state probabilities. Regime gating raises the excess-return Sharpe ratio from 0.55 to about 0.70 and cuts the maximum drawdown from -55% to between -33% and -17% . However, paired block-bootstrap intervals show that *no Sharpe difference is statistically distinguishable from zero at the 95% level*; the Sharpe edge concentrates in crisis eras, while the drawdown reduction is consistent across all eras. On a diversified SPY/TLT/GLD portfolio, no gate beats equal-weight buy-and-hold. The HMM never outperforms the volatility heuristic, and its filtered-state flicker triples turnover; a simple no-trade band repairs this, taking the HMM family from clearly worse than the diversified benchmark (Sharpe 0.64) to nearly matching it (0.79) with half the maximum drawdown. We conclude that regime detection is best understood as crash insurance, that model complexity must earn its costs and here does not, and that all findings are stable across bootstrap block lengths and walk-forward schedules.

1 Introduction and research question

Regime-switching ideas are pervasive in quantitative asset allocation: markets appear to alternate between calm and turbulent states, and conditioning exposure on the current state should, in principle, improve risk-adjusted outcomes. Published evidence, however, is often contaminated by look-ahead bias (smoothed rather than filtered state estimates), in-sample parameter selection, and ignored transaction costs.

RegimeLab is a research platform built to ask the question under deliberately conservative conditions. Three questions structure the study:

1. Does regime detection improve the *robustness* of simple strategies — out-of-sample, net of costs, with cash properly compensated — rather than their in-sample performance?
2. Does a statistical regime model (a Gaussian HMM) earn its complexity relative to a transparent volatility heuristic?
3. Do the answers survive diversification across asset classes, and can the cost side of regime trading be repaired mechanically?

The contribution is methodological as much as empirical: every defense against look-ahead bias is structural (encoded once, in one place, and tested), every experiment is reproducible from a configuration file and a seed, and every methodological decision is recorded in a public decision log.

2 Data

Prices. Daily adjusted closes from Yahoo Finance, cached immutably with JSON sidecars recording source and download date. The single-asset universe is SPY from January 2000; the multi-asset universe adds long-duration Treasuries (TLT) and gold (GLD) from December 2004 (GLD inception).

Cash rate. The 13-week T-bill yield index ($\text{\textasciitilde}IRX$), converted from an annualized percentage to a daily simple return as $y/100/252$ (a documented approximation), forward-filled over bond-market holidays on a continuous business-day calendar, and *lagged one day*: the rate known at the close of t accrues over $t+1$. Over the out-of-sample window the cash leg averaged about 1.7% per year.

Returns. Simple returns for portfolio aggregation; log returns for statistical modeling.

3 Methodology

3.1 Backtest protocol

Let w_t denote the vector of target portfolio weights decided at the close of day t (long-only, $\sum_i w_{i,t} \leq 1$). The engine applies a one-day execution lag — strategies never lag their own signals, so the lag exists in exactly one tested place:

$$h_t = w_{t-1}, \quad \tau_t = \sum_i |h_{i,t} - h_{i,t-1}|, \quad (1)$$

where h_t are the weights actually held during t and τ_t is the turnover. The net portfolio return with proportional costs c (in fractions per unit turnover) and cash rate r_t^f is

$$r_t^p = \sum_i h_{i,t} r_{i,t} + \max\left(0, 1 - \sum_i h_{i,t}\right) r_t^f - c \tau_t, \quad (2)$$

with the floor ensuring leveraged weights are never credited the cash rate. Costs use $c = 5$ basis points as the headline level and are swept over $\{0, 5, 20\}$ bps in every experiment. Sharpe and Sortino ratios are computed on returns in excess of r_t^f .

3.2 Regime models

All models implement one interface (fit on training prices; emit a causal regime label per date) and are fitted *per asset*.

Volatility threshold (heuristic). A date is labeled turbulent when trailing 20-day realized volatility exceeds the 80% quantile of the training sample’s volatility distribution; the quantile threshold is the only fitted state. Both parameters are swept in robustness checks.

Gaussian HMM. A K -state HMM with Gaussian emissions $r_t \mid s_t = k \sim \mathcal{N}(\mu_k, \sigma_k^2)$ on log returns, estimated by EM with several seeded restarts (best in-sample likelihood kept; convergence metadata recorded). Because EM state labels are arbitrary and change across refits, states are

reabeled by ascending variance, so state 0 is always the calmest. Crucially, prediction uses only the *filtered* distribution, computed by the forward recursion

$$\alpha_t(k) \propto \varphi(r_t; \mu_k, \sigma_k^2) \sum_{j=1}^K \alpha_{t-1}(j) A_{jk}, \quad P(s_t = k \mid r_{1:t}) = \alpha_t(k), \quad (3)$$

implemented in log space. Library routines based on the Viterbi path or on smoothed (forward-backward) probabilities condition on the full sample — i.e. on future data — and are used for diagnostics only. Truncation tests ($\text{predict}(x_{1:t}) = \text{predict}(x)_{1:t}$ for all t) enforce causality for every model.

3.3 Strategies

With $E(k) \in [0, 1]$ a fixed a-priori exposure map (flat in the highest-variance state, fully invested otherwise):

$$\text{hard gate: } w_t = b_t \cdot E(\hat{s}_t), \quad \hat{s}_t = \arg \max_k P(s_t = k \mid r_{1:t}) \quad (4)$$

$$\text{probability scaling: } w_t = b_t \cdot e_t, \quad e_t = \sum_k E(k) P(s_t = k \mid r_{1:t}) \quad (5)$$

$$\text{no-trade band: } w_t = \begin{cases} w_{t-1} & \text{if } |b_t e_t - w_{t-1}| \leq \delta \\ b_t e_t & \text{otherwise,} \end{cases} \quad (6)$$

where b_t is the base strategy's weight (buy-and-hold or a 200-day moving-average trend rule). The band recursion is forward-only and passes the same truncation tests. The band level δ is *reported as a sweep* over $\{0.05, 0.10, 0.20\}$, never selected in-sample.

3.4 Walk-forward validation

Models are refitted on an expanding window: an initial training sample of 1,260 trading days (≈ 5 years), refit every 252 days. Only out-of-sample labels and probabilities — stitched across refits — ever reach a strategy. This yields 22 refits on SPY (out-of-sample 2005-01-07 to 2026-06-09, 5,388 days) and 17 refits on the multi-asset universe (2009-12-02 to 2026-06-09, 4,154 days). The out-of-sample window and cost levels are identical for gated and ungated strategies, so differences are attributable to the regime signal alone.

3.5 Statistical inference

A point difference between two Sharpe ratios is not a finding. We use a *paired circular block bootstrap*: indices are resampled in contiguous 21-day blocks (wrapping at the boundary), the same blocks are applied to both return series (preserving cross-correlation), and 95% percentile intervals are formed over 2,000 draws. We also report $P(\leq 0)$, the fraction of bootstrap draws in which the Sharpe difference is non-positive — a descriptive bootstrap quantity, not a formal p -value. Block length is itself swept over $\{5, 21, 63\}$ days as a nuisance check.

4 Implementation

The repository separates an installable, unit-tested library (`src/regimelab/`: data, features, regimes, strategies, backtest engine, validation, inference, plotting) from configuration-driven experiments (`experiments/` + `configs/`); notebooks only load experiment outputs and call shared

plotting helpers. The test suite (102 tests) includes dedicated look-ahead defenses: a clairvoyant-signal canary for the engine lag, truncation causality tests for every feature and regime model, and train-precedes-test assertions for the walk-forward splits. A regression guard verified that the v3 additions (cash leg, bands) reproduce earlier results exactly when disabled. All methodological decisions and their rationale are recorded in `reports/decisions.md`.

5 Results: single asset (SPY)

Table 1 reports the out-of-sample headline results at 5 bps. Figures 1 and 2 show the corresponding wealth and drawdown paths.

Table 1: SPY, out-of-sample 2005–2026, net of 5 bps, cash leg active. Sharpe and Sortino are on excess returns; turnover is one-way, annualized. `hmm2p_bXX` denotes probability scaling with a no-trade band $\delta = 0.XX$.

Strategy	Ann. ret.	Ann. vol.	Sharpe	Sortino	Max DD	Turnover
buy_and_hold	10.9%	19.0%	0.55	0.78	−55.2%	0.05×
trend_200d	8.3%	11.6%	0.60	0.81	−20.9%	6.3×
vol20q80_bh	10.3%	12.8%	0.70	0.95	−33.2%	3.4×
vol20q80_trend	8.5%	11.1%	0.64	0.87	−20.9%	5.9×
hmm2_bh	9.3%	11.4%	0.69	0.95	−21.4%	9.4×
hmm2_trend	8.2%	10.3%	0.65	0.89	−17.2%	7.9×
hmm2p_bh	8.8%	10.8%	0.67	0.93	−22.0%	11.2×
hmm2p_b05_bh	8.8%	10.7%	0.68	0.93	−22.0%	10.0×
hmm2p_b10_bh	8.6%	10.5%	0.68	0.93	−21.0%	8.8×
hmm2p_b20_bh	8.8%	10.1%	0.71	0.99	−16.8%	6.7×
hmm3_bh	10.1%	17.1%	0.55	0.76	−51.8%	4.1×

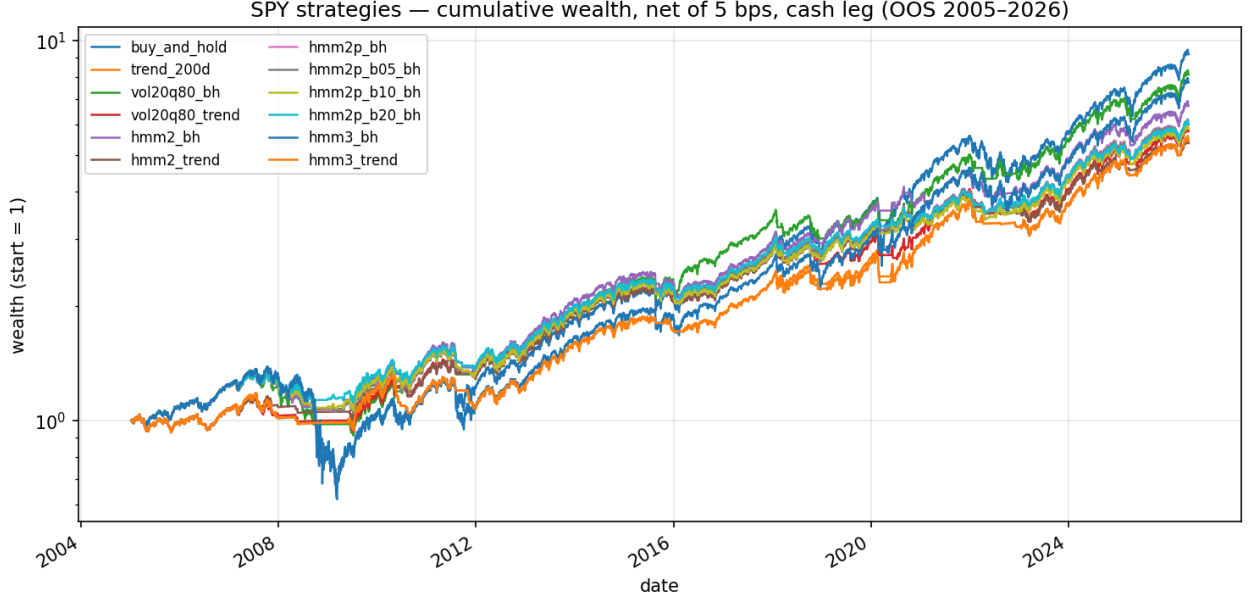


Figure 1: Cumulative wealth, net of 5 bps with the cash leg, out-of-sample 2005–2026 (log scale). Gated variants track buy-and-hold in calm markets and step aside in turbulence: comparable terminal wealth with visibly shallower paths through 2008–09, 2020, and 2022. The 3-state HMM gate (`hmm3_bh`) barely deviates from buy-and-hold because its extreme state fires on only 2.5% of days.

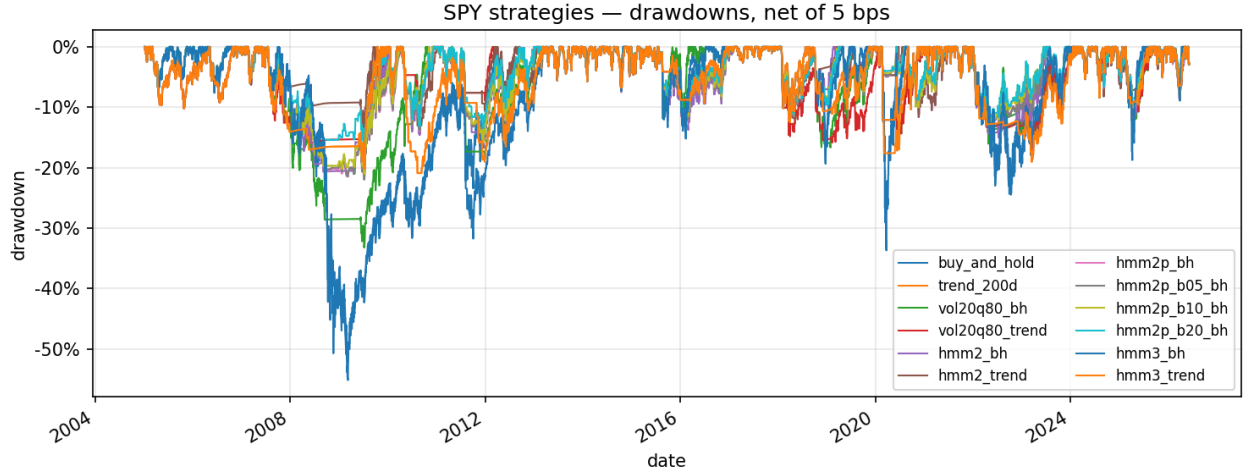


Figure 2: Drawdowns net of 5 bps. The headline effect of regime gating is visible here rather than in terminal wealth: maximum drawdown improves from -55% (buy-and-hold) to -33% (volatility gate) and -17 to -22% (HMM variants).

5.1 What the HMM detects

Figure 3 overlays the out-of-sample turbulent labels (21.0% of days; the volatility filter flags 14.8%) on the SPY price, with the filtered turbulence probability below. Labels concentrate on 2008–09, 2011, 2015–16, 2018, March 2020, and 2022. Near regime boundaries the filtered probability oscillates around 0.5; each oscillation is a round trip for a hard-gated strategy. This *filtered-state flicker* is the mechanism behind the HMM’s $9.4\times$ annual turnover and is the target of the no-trade

band.

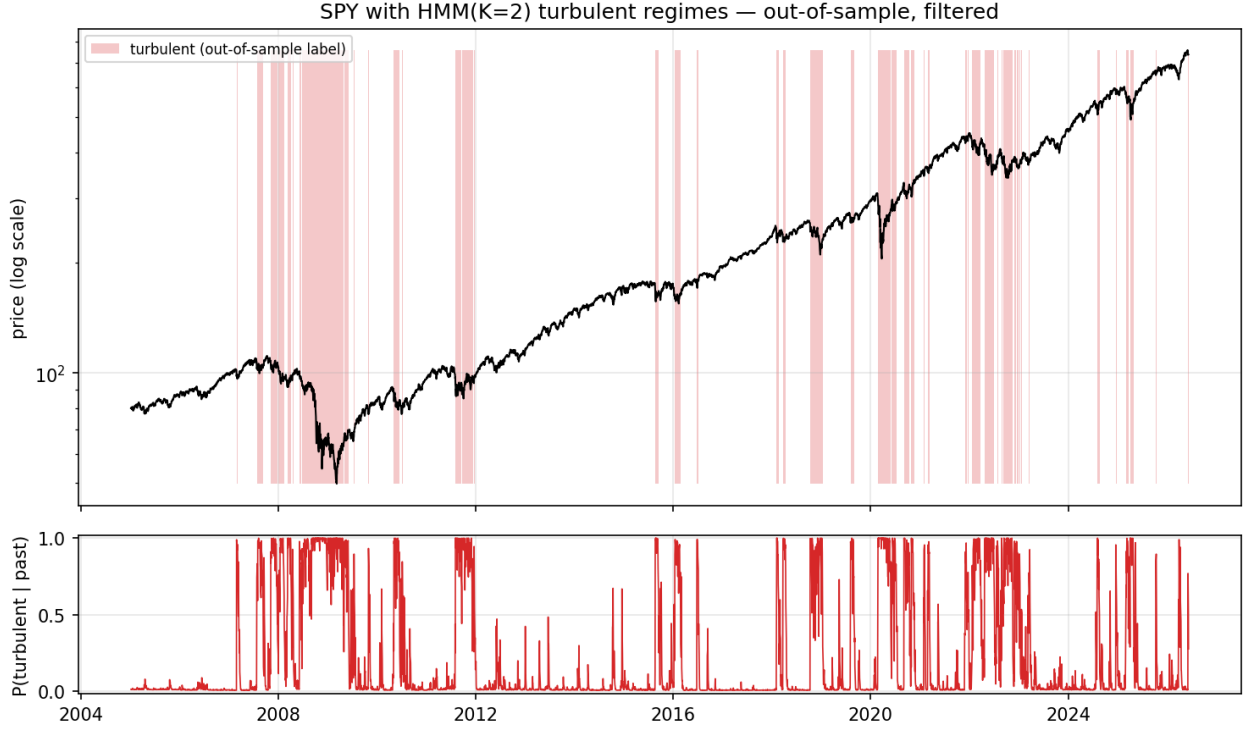


Figure 3: SPY (log scale) with out-of-sample HMM($K=2$) turbulent regimes shaded (top) and the filtered probability of the turbulent state (bottom). Estimates at date t use information up to t only. The probability panel shows the boundary flicker that drives turnover in hard-gated strategies.

5.2 Banding the flicker

The band sweep cuts turnover monotonically ($11.2\times \rightarrow 10.0\times \rightarrow 8.8\times \rightarrow 6.7\times$) at no Sharpe cost; the widest band has the HMM family’s best Sharpe (0.71) and best maximum drawdown (-16.8%) (Figure 4). We read this as a sweep result, not a tuned point: the scaled strategy’s Sharpe is flat in δ while its trading costs are not, so banding is weakly dominant.

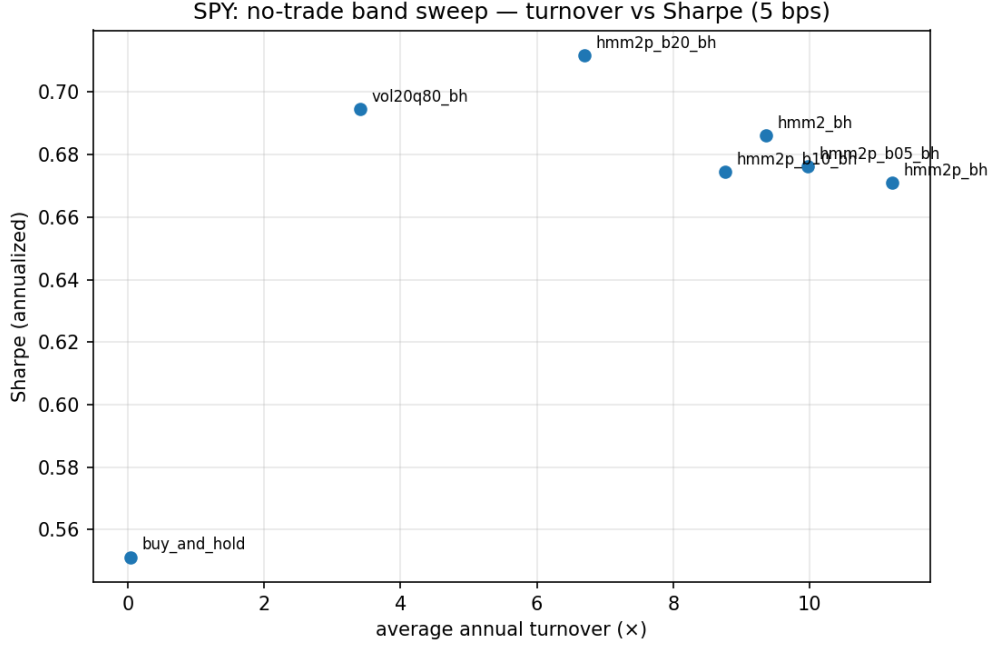


Figure 4: Turnover versus Sharpe (SPY, 5 bps). Moving from the unbanded probability-scaled strategy (`hmm2p_bh`, right) through increasing band levels traces an up-and-left path: less trading, equal or better risk-adjusted performance.

5.3 Is any of it statistically significant?

Table 2: Paired circular block bootstrap (21-day blocks, 2,000 draws) on excess-return Sharpe differences, SPY at 5 bps. $P(\leq 0)$ is the bootstrap fraction of non-positive differences (descriptive).

Comparison	ΔSharpe	95% CI	$P(\leq 0)$
vol20q80_bh vs. buy_and_hold	+0.14	$[-0.15, +0.46]$	0.19
hmm2_bh vs. buy_and_hold	+0.14	$[-0.18, +0.48]$	0.24
hmm2p_b20_bh vs. buy_and_hold	+0.16	$[-0.15, +0.49]$	0.18
hmm3_bh vs. buy_and_hold	-0.01	$[-0.15, +0.16]$	0.57
vol20q80_bh vs. hmm2_bh	+0.01	$[-0.21, +0.22]$	0.48
hmm2_bh vs. hmm2p_bh	+0.02	$[-0.08, +0.11]$	0.38

Every 95% interval includes zero (Table 2, Figure 5): twenty-one years of daily data cannot certify a Sharpe improvement of ≈ 0.15 . The volatility heuristic and the 2-state HMM are statistically indistinguishable from each other.

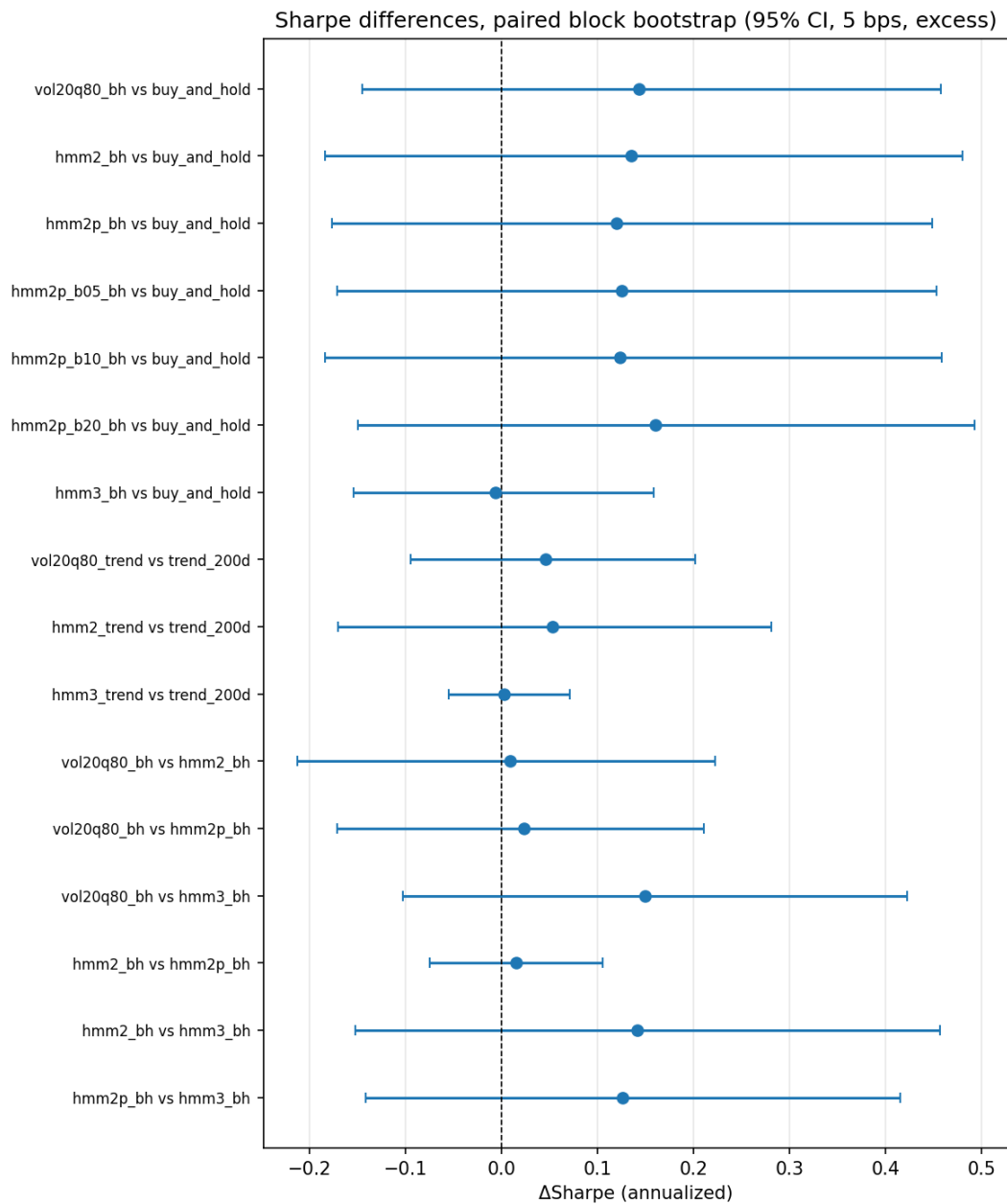


Figure 5: Sharpe differences with 95% paired-bootstrap intervals (SPY, 5 bps, excess returns). All intervals straddle zero.

5.4 Sub-periods, parameters, and nuisance choices

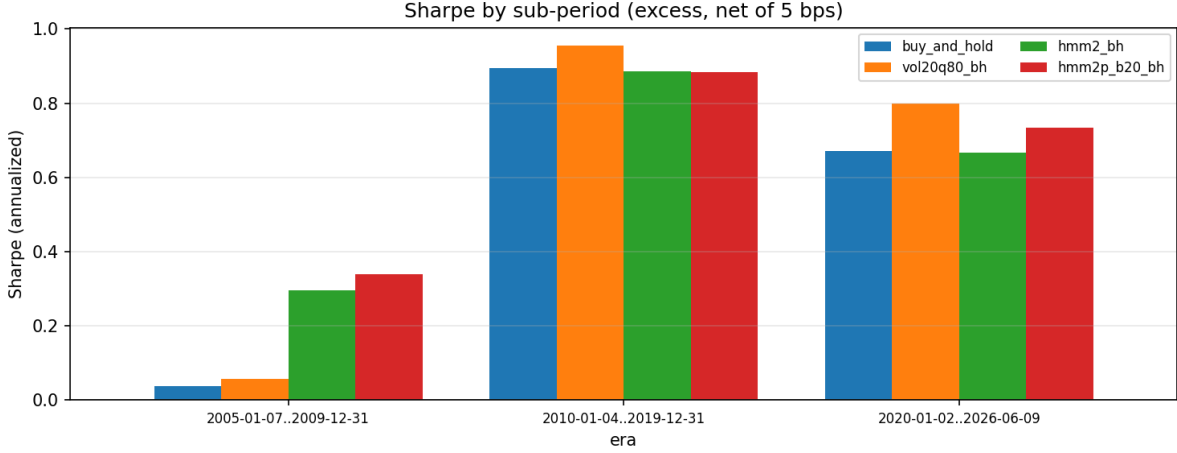


Figure 6: Excess Sharpe by sub-period. The Sharpe edge of gating lives in the 2005–09 crisis era (e.g. buy-and-hold 0.04 vs. `hmm2_bh` 0.30) and partially in 2020+; in the calm 2010s gating neither helped nor hurt. The drawdown reduction (not shown) appears in every era.

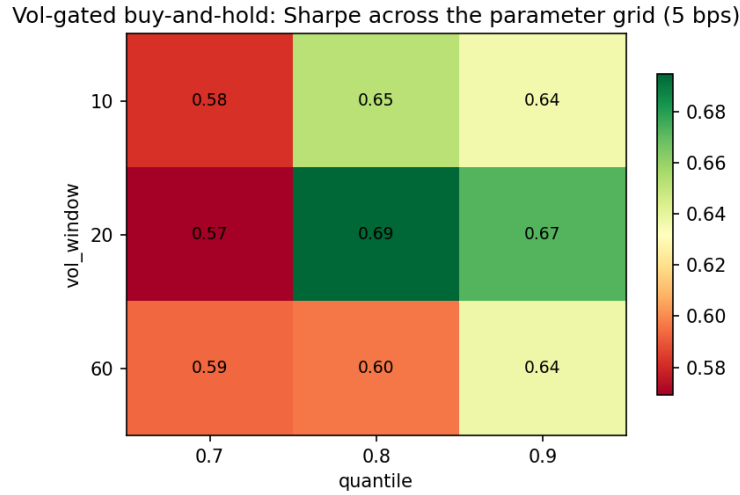


Figure 7: Volatility-gate parameter grid (Sharpe of gated buy-and-hold, 5 bps). A plateau, not a peak: every cell beats buy-and-hold’s 0.55, so the headline configuration (20d, $q=0.8$) is not a tuned artifact.

Two nuisance checks complete the picture. Across bootstrap block lengths $\{5, 21, 63\}$, $P(\leq 0)$ for the headline comparisons moves by less than 0.03 (e.g. volatility gate vs. buy-and-hold: 0.18/0.19/0.18). Across walk-forward schedules ($\{\text{expanding, rolling}\} \times \{126, 252\}$ -day refits), the volatility gate’s Sharpe stays within 0.70–0.78. No conclusion hinges on either constant.

6 Results: multi-asset (SPY/TLT/GLD)

Each asset is gated by its own independently fitted regime model; the baseline is the equal-weight portfolio. Table 3 and Figures 8–10 report the results.

Table 3: Multi-asset universe (equal-weight SPY/TLT/GLD), out-of-sample 2010–2026, net of 5 bps, cash leg active.

Strategy	Ann. ret.	Ann. vol.	Sharpe	Max DD	Turnover
vol20q80_bh	7.5%	7.5%	0.82	−12.7%	5.0×
buy_and_hold	8.9%	9.3%	0.81	−22.7%	0.06×
hmm2p_b10_bh	6.1%	5.9%	0.79	−10.2%	4.9×
hmm2p_b05_bh	6.2%	6.5%	0.74	−12.1%	7.4×
hmm2p_bh	6.3%	6.8%	0.72	−12.2%	11.3×
vol20q80_trend	5.7%	6.1%	0.72	−13.3%	9.2×
hmm2p_b20_bh	5.0%	5.1%	0.71	−8.3%	2.4×
hmm2_bh	6.0%	7.3%	0.64	−15.0%	9.5×
trend_200d	5.8%	7.2%	0.63	−12.6%	8.4×

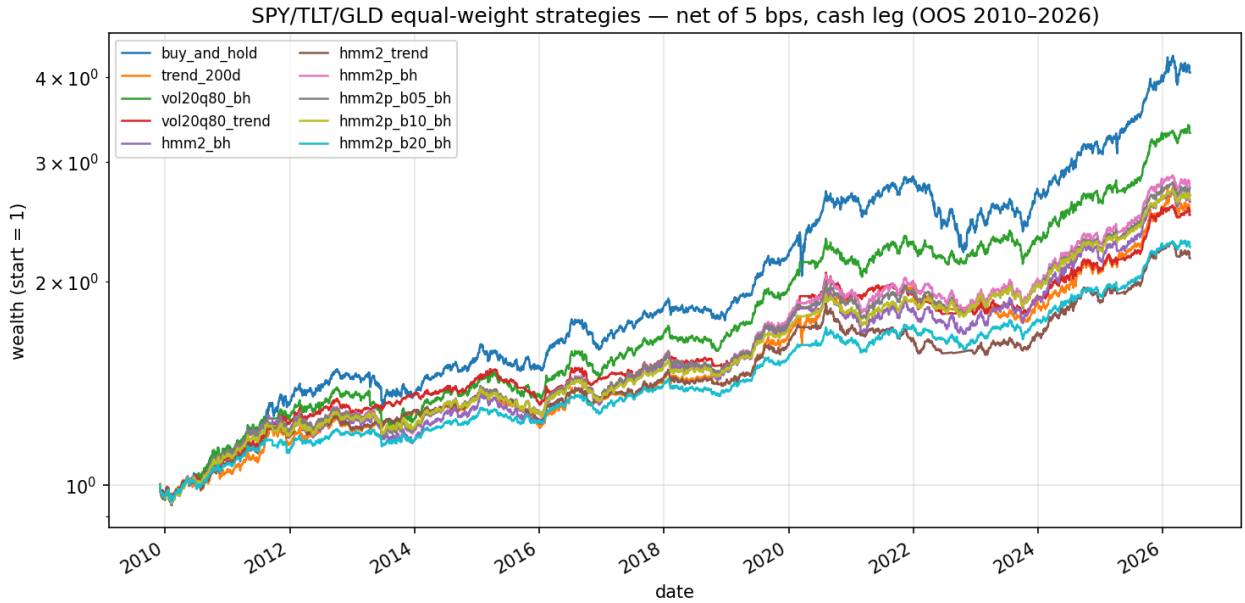


Figure 8: Multi-asset cumulative wealth, net of 5 bps with the cash leg. The equal-weight portfolio is already smooth; gated variants trade return for further drawdown reduction.

The headline result is negative and informative: the diversified equal-weight portfolio achieves excess Sharpe 0.81 with a −22.7% maximum drawdown, and no gated variant improves on it with any confidence (volatility gate: $\Delta\text{Sharpe} +0.01$, $P(\leq 0) = 0.46$; hard HMM gate: -0.16 , $P(\leq 0) = 0.87$). The strongest bootstrap signals in the study are orderings *within* the gated family: the volatility heuristic beats the hard HMM gate ($+0.17$, $P(\leq 0) = 0.04$), and soft scaling beats hard gating ($P(\text{hard} \geq \text{soft}) = 0.045$).

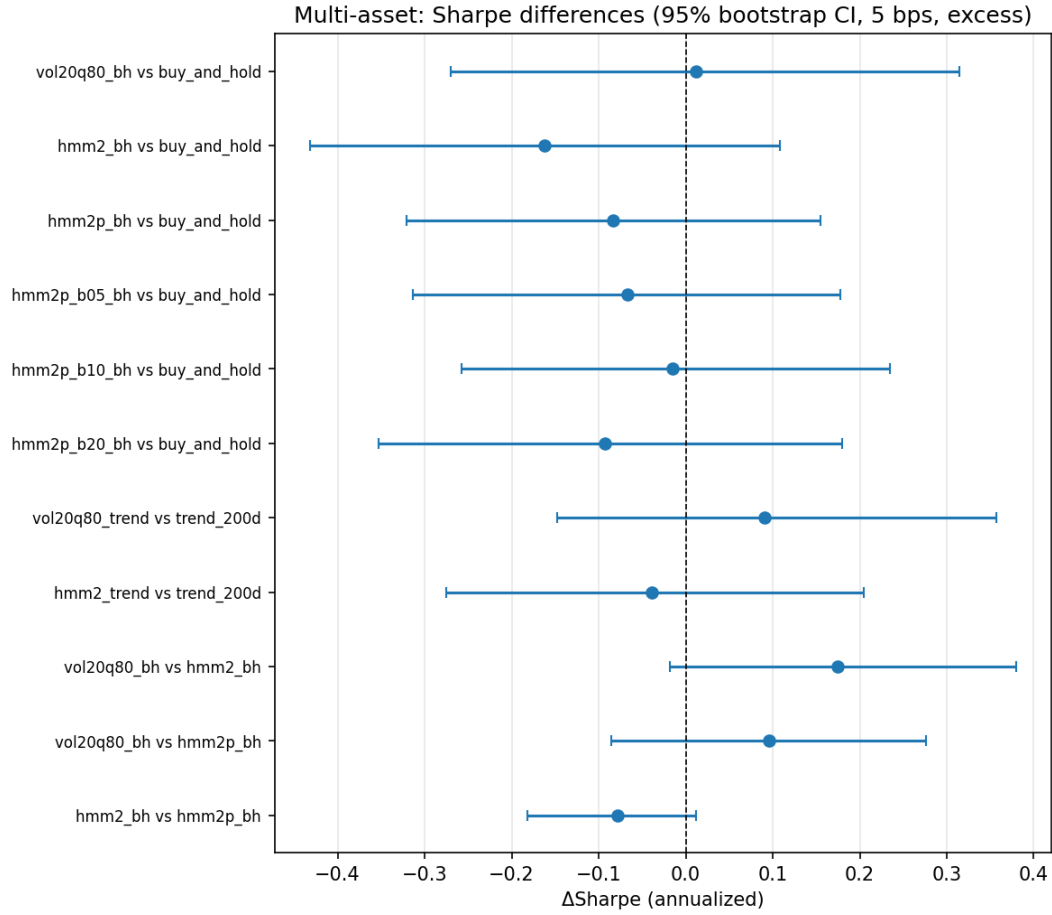


Figure 9: Multi-asset Sharpe differences with 95% paired-bootstrap intervals. Nothing beats diversified buy-and-hold; the informative orderings are within the gated family (heuristic > hard HMM; soft > hard).

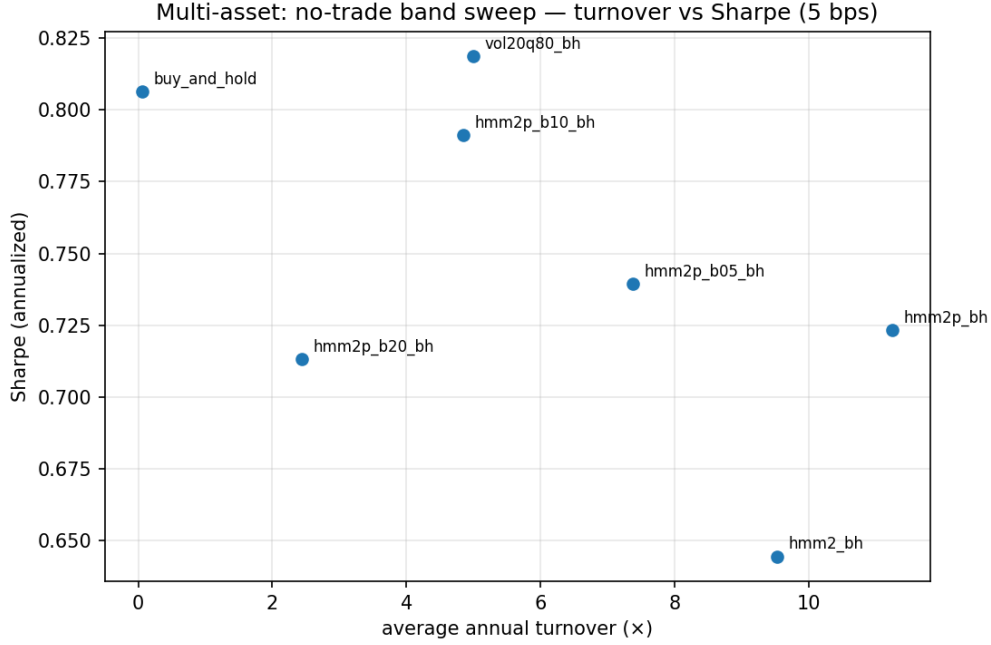


Figure 10: Multi-asset turnover-Sharpe frontier. The band sweep moves the HMM family from clearly worse than buy-and-hold (hard gate: Sharpe 0.64 at $9.5\times/\text{yr}$) to nearly matching it: $\delta=0.10$ gives Sharpe 0.79 with half the drawdown (-10% vs. -23%) on $4.9\times/\text{yr}$ turnover; $\delta=0.20$ cuts the drawdown to -8% on $2.4\times/\text{yr}$.

7 Discussion and conclusions

1. **Regime gating is crash insurance.** In every configuration tested it cuts deep drawdowns at some cost in return, and the drawdown improvement — unlike the Sharpe improvement — appears in every sub-period. Sharpe gains of ≈ 0.15 never clear paired bootstrap scrutiny on 21 years of daily data; claims of regime-based Sharpe enhancement at this sample size deserve suspicion.
2. **Complexity did not pay.** The HMM never beat the volatility heuristic where both worked (SPY: $\Delta\text{Sharpe} +0.01$) and was clearly worse in the multi-asset setting ($P(\leq 0) = 0.04$ for the heuristic-HMM comparison), where it trades bond and gold volatility states that are not reliably associated with negative returns the way equity volatility is.
3. **Diversification first.** The cheapest “regime model” in the study is holding three lowly-correlated assets: equal-weight SPY/TLT/GLD matches the best gated single-asset Sharpe at near-zero turnover. This is established here with a proper cash leg, so it is not an artifact of uncompensated cash.
4. **If using a probabilistic model: scale, band, do not threshold.** The ordering hard gate $<$ soft scaling $<$ banded scaling holds in both universes. Banding fixes the cost fragility that made the HMM strictly worse at 20 bps, and multi-asset it converts the HMM from a detractor into cheap insurance.
5. **The findings are not nuisance-parameter artifacts** — stable across bootstrap block lengths and walk-forward schemes and cadences.

8 Limitations

The cash conversion uses the discount-yield approximation $y/100/252$; negligible at historical yields, but documented rather than eliminated. The band levels were swept, not optimized, yet the sweep grid itself was chosen by the authors; a pre-registered grid would be cleaner. Multi-asset gating is per-asset and ignores cross-asset information. The data source is Yahoo Finance, and the multi-asset sample covers roughly one and a half market cycles. Finally, with about a dozen strategy variants, multiple-comparison effects deserve formal treatment (e.g. deflated Sharpe ratios) once the strategy set stops growing.

9 Next steps

1. A *joint multivariate* HMM on the SPY/TLT/GLD return vector, against the per-asset models used here — testing whether cross-asset structure rescues the statistical approach.
2. A deflated-Sharpe / multiple-comparison appendix.
3. Block-bootstrap inference on drawdown-based metrics, the quantity regime gating actually improves.

A Reproducibility

All results in this report regenerate from a clean checkout with:

```
pip install -e ".[models,data,notebooks,dev]"
pytest                                     # 102 tests
python experiments/run_regime_comparison.py configs/comparison_spy.yaml
python experiments/run_robustness.py       configs/comparison_spy.yaml
python experiments/run_regime_comparison.py configs/multiasset.yaml
python experiments/run_robustness.py       configs/multiasset.yaml
PYTHONPATH=src jupyter nbconvert --to notebook --execute \
    --inplace notebooks/0*.ipynb
```

Experiment outputs land in `experiments/outputs/`, one directory per experiment; the figures used here are produced by the notebooks into `reports/figures/`. The methodological decision log is `reports/decisions.md`; this document is compiled with `make -C reports` (Tectonic).